AINK – AI-FIRST E-INK DISPLAY

by UkieLab

ABOUT

The AInk by UkieLab is a 7.5" e-ink display designed for AI agents. Once connected to your WiFi, it runs a persistent HTTP server that AI assistants can discover via mDNS or IP address and control by sending JSON drawing commands. The 800x480 black-and-white display retains its image even when powered off. The device stays always-on with WiFi active, ready to receive commands.

AI agents can draw text, shapes, images, QR codes, and more on the display. The device also serves an OpenAPI spec and MCP tool definitions for easy AI integration.

SETUP

Let's begin the setup process:

1. **Connect to the device "UkieLab-AInk" WiFi**
    a. Scan the QR code shown on the device screen to connect automatically or on your phone, go to Settings > WiFi and look for "UkieLab-AInk"
    b. Once connected, a configuration page will open automatically in your browser. If it does not, scan QR code from Step 2 or open http://192.168.4.1 in your browser
2. **Configure your device**
    a. On the configuration page, give your device a name (this becomes its mDNS hostname). Then select your home WiFi network from the list and enter your WiFi password.

AInk connects to your WiFi and displays a startup screen showing its IP address and a QR code. Your AI assistant can now discover the device via mDNS or you can point it to the IP address shown on screen. Ask your AI to visit the device URL to get started.

USING AINK WITH AI ASSISTANTS

Once your AInk device is on the network, any AI assistant on the same WiFi can control it. Here is how to get started:

1. **Tell your AI about the device**
    a. Give your AI assistant the device's IP address (shown on the display after setup) and ask it to read the API and MCP documentation
2. **Ask your AI agent to show you something useful**
    a. The AI can draw directly on a device, and it can also check what is displayed. It might take a few itterations for your agent to learn how to use it. Once you have something you like ask your agent to turn it into skill and show you on a schedule (this will depend on a setup you have).
3. **All set!**

QUESTIONS OR SUGGESTIONS?

Contact support at **ukielab.com** or ukielab@gmail.com

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F.A.Q.

How do I reset to factory settings?

Press the Reset button 3 times within 10 seconds. This clears WiFi credentials and device configuration. After the reset, the device will restart in setup mode and display the setup screen with instructions to connect to the "UkieLab-AInk" WiFi network.

How do I resolve any issues with rendering

Device is fully controlled by AI agent and in case something is not displayed as it should you have to ask AI to check it and fix it. AI agents like Claude Code are able to check what's on display, identify most of the rendering issues and fix them.

How do I change the device name?

While the device is connected to WiFi, it displays its IP address on the startup screen. Open that address in a browser on a phone or laptop connected to the same WiFi.

How do I update the WiFi password?

Reset to factory settings by pressing the Reset button 3 times within 10 seconds. The device will restart in setup mode. Connect to the "UkieLab-AInk" WiFi and follow the regular setup process to enter the new WiFi credentials.

How do I update the firmware?

The device supports over-the-air (OTA) firmware updates from GitHub releases. Visit the device's IP address in a browser and navigate to the update page. You can check for updates and install them directly from the web interface. If auto-update is enabled, the device checks for updates nightly at 2 AM.

How do AI assistants discover the device?

The device advertises itself via mDNS with the service type _aiscreen._tcp on port 80. AI assistants can discover it automatically on the local network. The mDNS TXT records include the device model, firmware version, display dimensions, and API version. You can also provide the device IP address directly to your AI assistant.

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